Turtle Games: Technical Report

Background and Business Context

Turtle Games is a global manufacturer and retailer of games and related products, including books, board games, video games, toys, and third-party items. The organisation has collected sales data and customer review data to better understand customer behaviour. A key strategic objective is to improve sales performance by analysing customer engagement trends, particularly around the company's loyalty programme.

This report will address loyalty point accumulation, the potential for customer segmentation and targeted marketing, how customer reviews can inform marketing and product improvements, and the suitability of loyalty points data for predictive modelling.

Analytical Approach

Analysis was completed in both *Python* and *R* to align with departmental preferences, however data transformation methods remained consistent between the two.

The dataset contained 2,000 observations with demographics (gender, age, education), numeric measures (remuneration, spending score, loyalty points, product) and text fields (review, summary). Redundant fields were removed (language, platform).

Data preparation followed a reproducible method of structural checks, standardisation and cleaning. This ensured continuity across analysis and reduced the risk of errors.

Python modelling used *statsmodels OLS* to quantify linear relationships between loyalty points and types of observations, with coefficients, standard errors and predicted values extracted for regression tables and plots. Spending score and remuneration showed strong positive relationships with loyalty points, while age exhibited a weaker effect.

A *DecisionTreeRegressor (sklearn)* was fit to explore non-linear thresholds, with pruning applied (maximum depth and minimum samples per leaf) to improve interpretability and limit overfitting. This revealed clear behavioural thresholds, particularly around spending score, reinforcing findings from regression while highlighting limitations of linear assumptions.

For segmentation, k-means clustering was applied to remuneration and spending score. Features were scaled before clustering to ensure comparability between spending score and remuneration. The Elbow and Silhouette methods both indicated that a five-cluster solution would be optimal, resulting in clear, actionable customer segments based on spending and income profiles.

For text analytics, only the text fields *review* and *summary* were retained; text was lowercased, punctuation removed, duplicates dropped, tokenised, and analysed via frequency distributions and polarity scoring. This allowed identification of dominant positive and negative themes and extraction of the most extreme positive and negative reviews.

In R, the exploratory analysis was repeated using *readr* and *ggplot2* to recreate key histograms, boxplots, and scatterplots, while *summary()* and *skimr* were used to sense-check the data. Additional focus was placed on distribution diagnostics using Q–Q plots and the Shapiro–Wilk test.

The final multiple linear regression model specified loyalty points as the target with spending score, remuneration and age as numeric predictors. Residual diagnostics were reviewed to validate assumptions and support scenario-based prediction. This provided organisationally aligned statistical validation in the Sales teams preferred analytical environment.

Visualisation and Insights

Visualisations and analytical outputs were developed progressively to address Turtle Games's core business questions around loyalty engagement, segmentation, and marketing effectiveness.

Initial exploratory analysis focused on how customers accumulate loyalty points using scatterplots. Visualising loyalty points against spending score, remuneration, and age revealed clear patterns: spending score showed the strongest positive relationship, remuneration a weaker but still positive association, and age little visible correlation. Loyalty points ranged from 25 to 6,847, with increasing dispersion at higher spending and income levels, indicating heterogeneous behaviour among high-value customers.

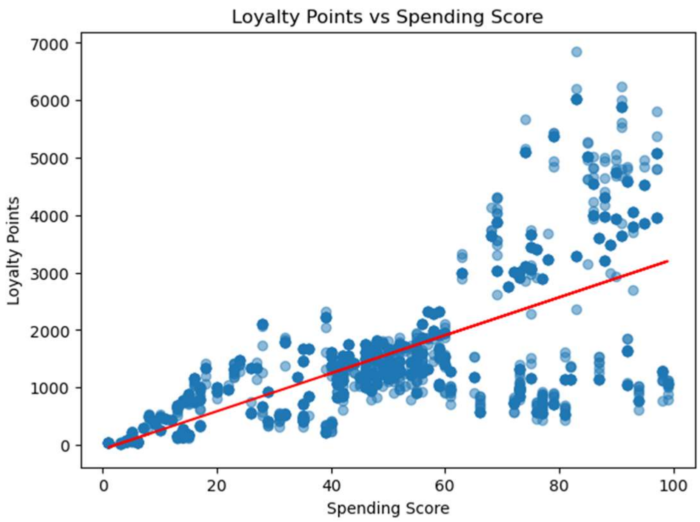

Figure 1 - Loyalty Points vs. Spending Score

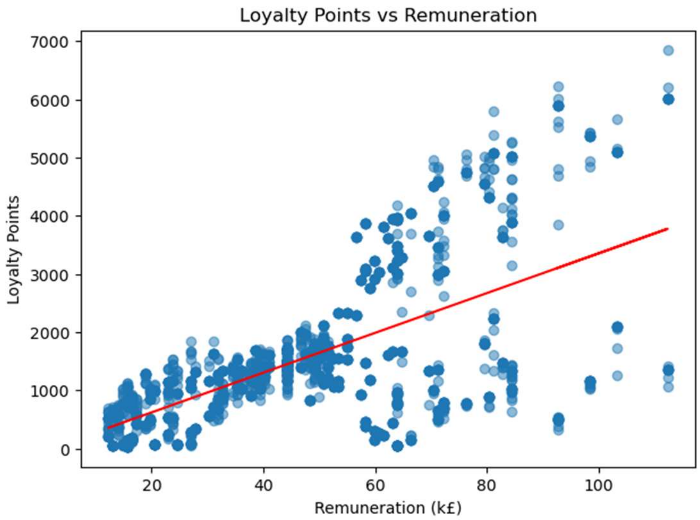

Figure 2 - Loyalty Points vs. Remuneration

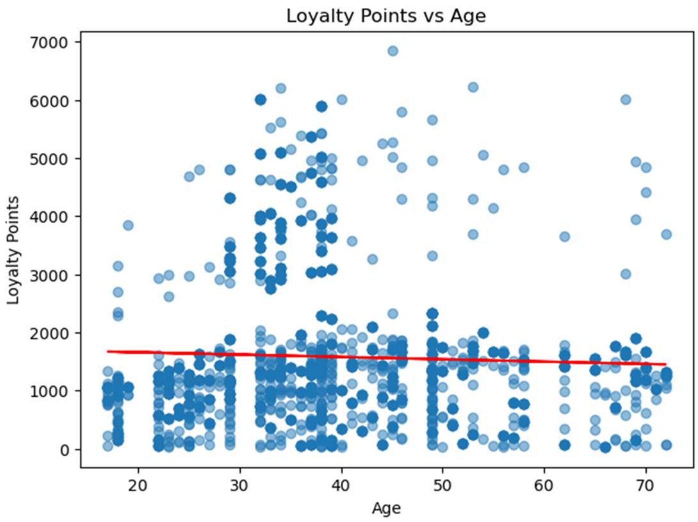

Figure 3 - Loyalty Points vs. Age

Decision tree analysis was used to explore the structure underlying this behaviour. Tree visualisation identified spending score as the root split, confirming it as the dominant driver of loyalty accumulation.

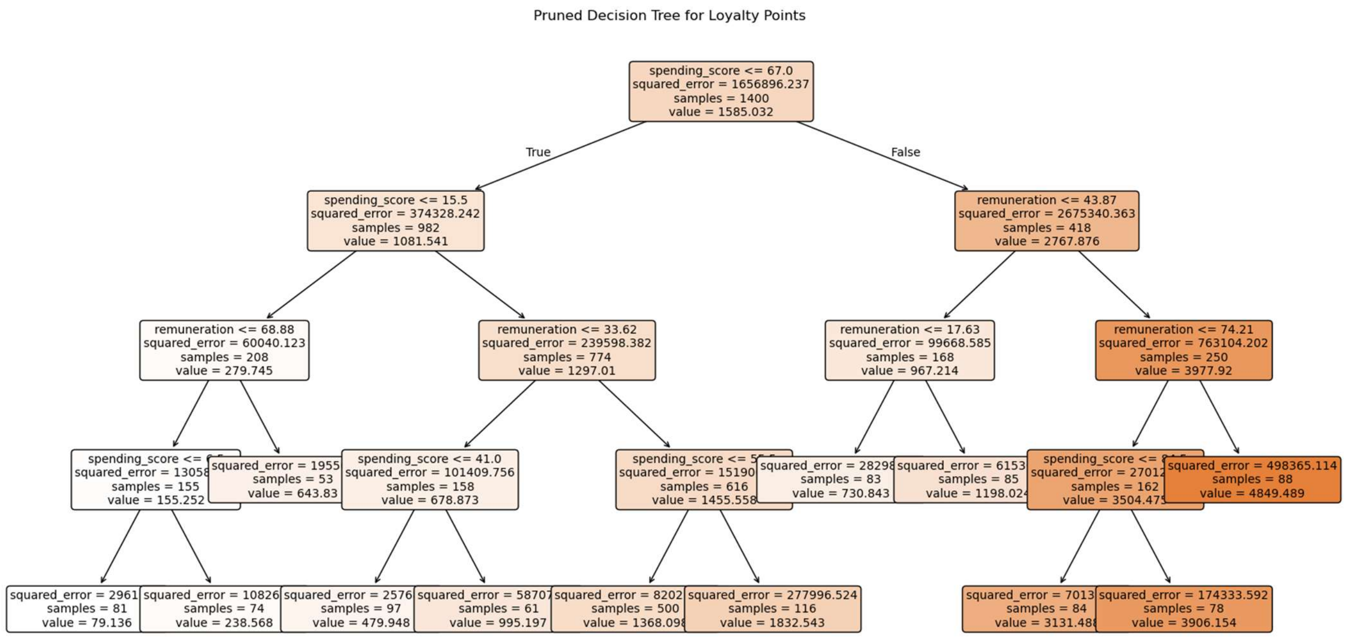

Figure 4 - Pruned Decision Tree for Loyalty Points

Customers with spending scores below approximately 65–70 accumulated substantially fewer loyalty points, while outcomes diverged sharply above this threshold. Secondary splits on remuneration highlighted further non-linear effects among high spenders, reinforcing that income moderates—but does not replace—spending behaviour. These rule-based visuals provided intuitive insight into behavioural thresholds that linear models alone could not capture.

Analysis was extended through k-means clustering. After scaling features, Elbow and Silhouette diagnostics indicated that five clusters offered the best balance between cohesion and separation (silhouette ≈ 0.58).

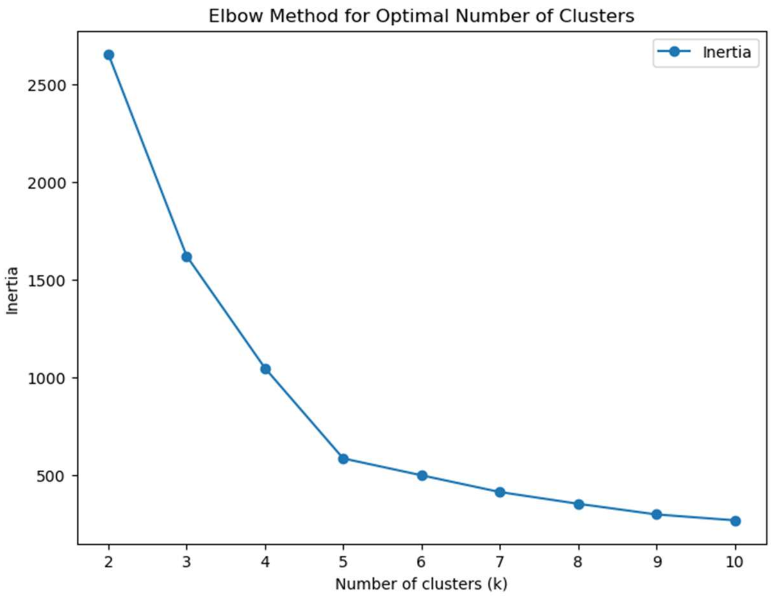

Figure 5 - Elbow Method

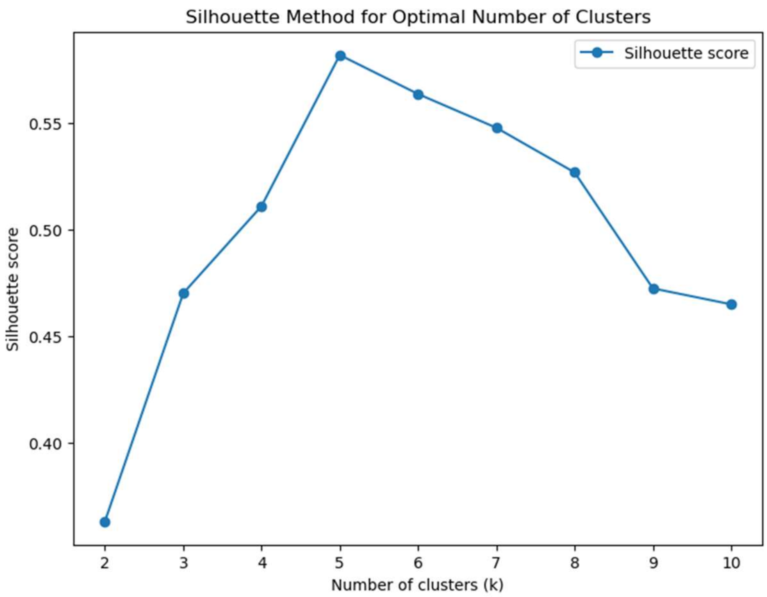

Figure 6 - Silhouette Method

Cluster sizes ranged from approximately 270 to 770 customers, with one dominant mainstream segment and several smaller, high-value clusters characterised by higher spending and income. These visual cluster separations demonstrated clear opportunities for differentiated marketing and loyalty incentives rather than uniform programme design.

Figure 7 - K-Means Clustering of Customers

Customer reviews were analysed using NLP techniques. Frequency analysis showed that positive terms such as *"game," "great,"* and *"fun"* dominated both reviews and summaries. Polarity analysis indicated overall positive sentiment, with mean polarity scores of approximately 0.21 for reviews and 0.22 for summaries. Summaries exhibited greater dispersion (standard deviation ≈ 0.34 vs 0.26), indicating more extreme sentiment in short-form text. Negative extremes repeatedly referenced usability issues, poor value, and complicated instructions, providing directly actionable insights.

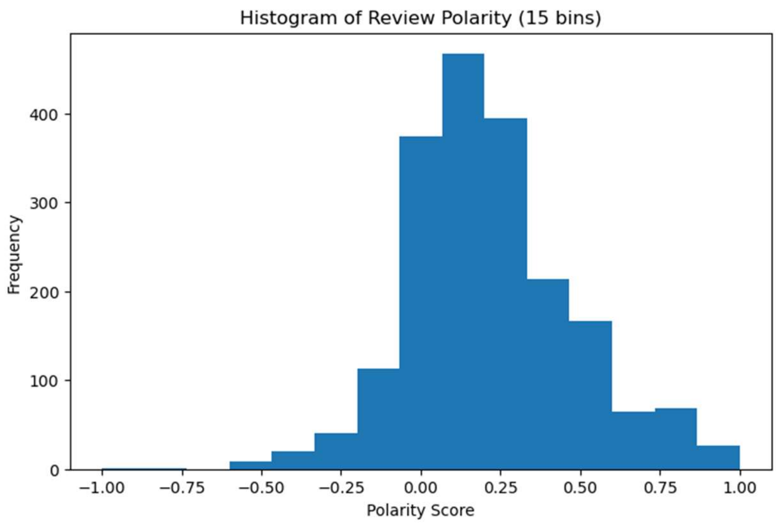

Figure 8 - Histogram of Review Polarity

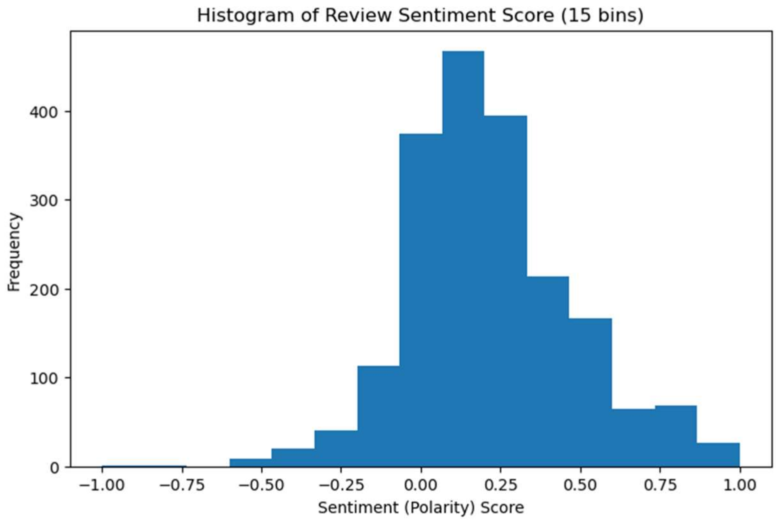

Figure 9 - Histogram of Review Sentiment Score

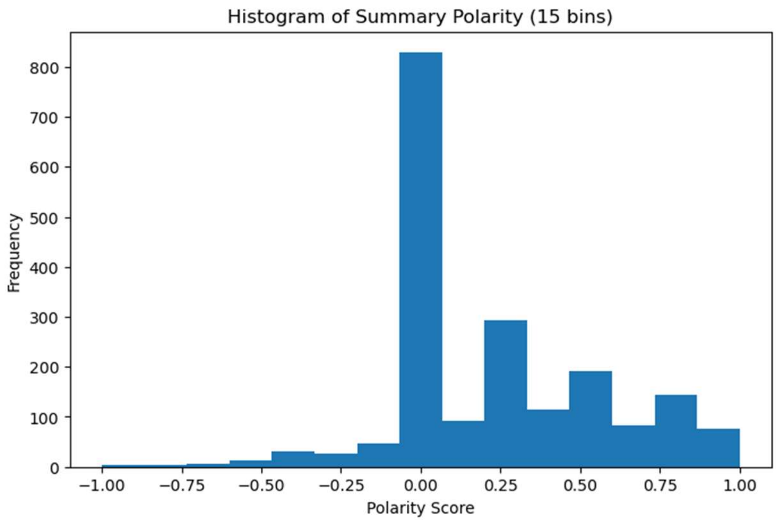

Figure 10 - Histogram of Summary Polarity

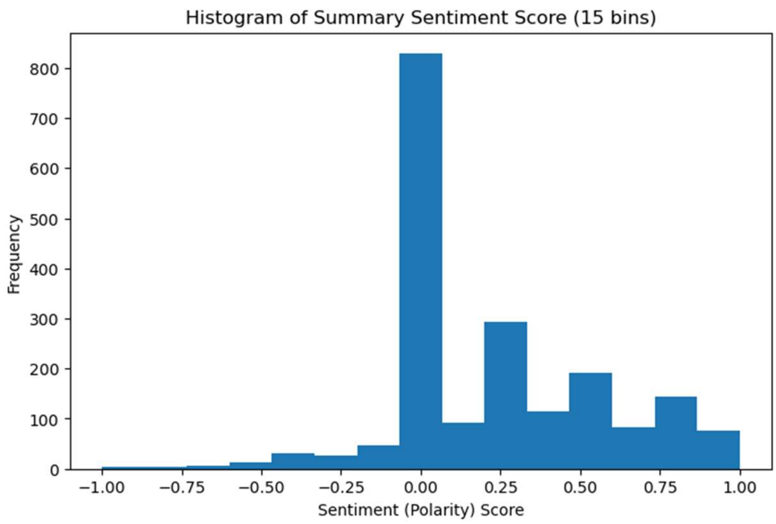

Figure 11 - Histogram of Summary Sentiment Score

The *Python* analysis was replicated in *R* using *tidyverse* and *ggplot2*, introducing histograms and boxplots to confirm the right-skewed distribution of loyalty points.

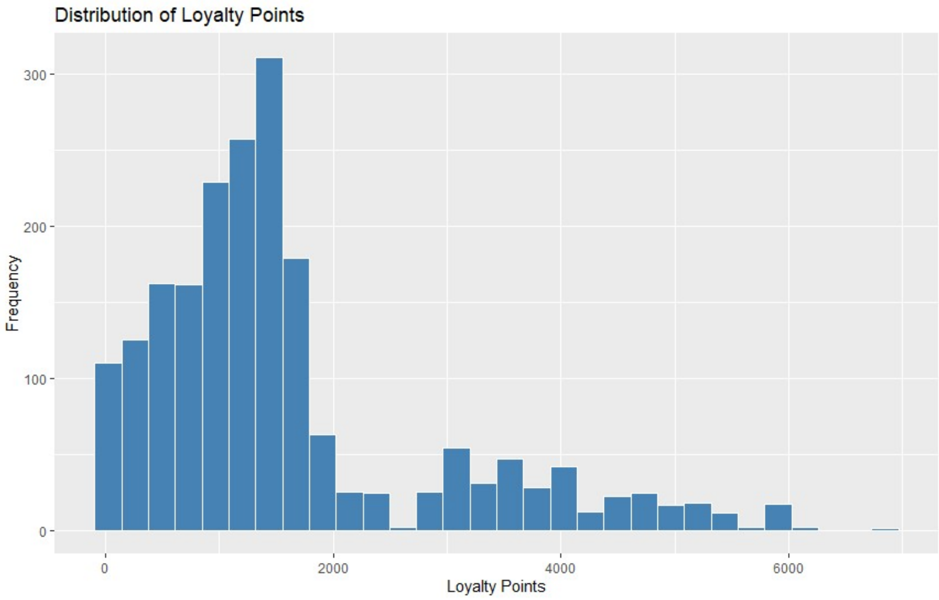

Figure 12 - Loyalty Point Distribution Histogram

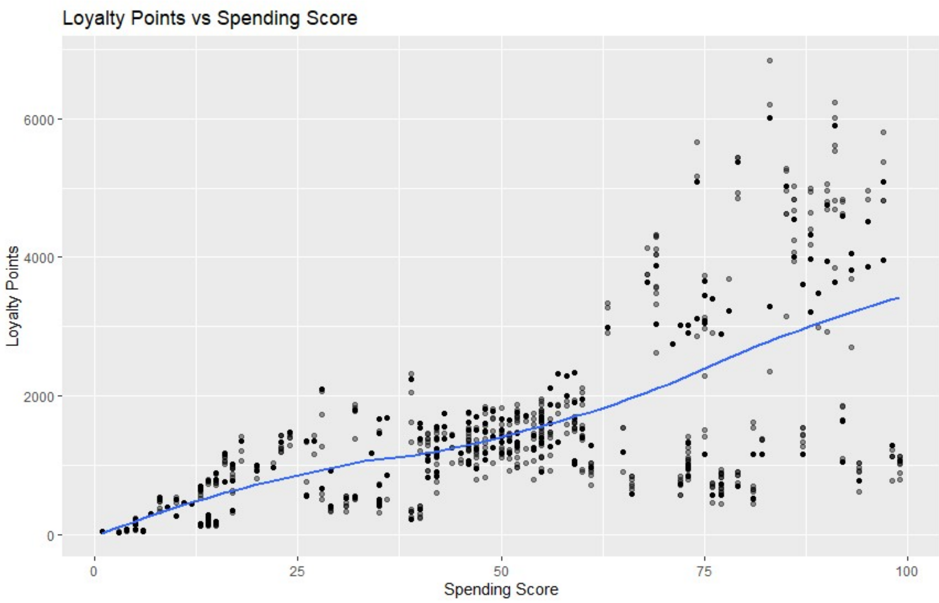

Figure 13 - Replication in R of earlier scatterplot

Q–Q plots and Shapiro–Wilk test ($W \approx 0.84$, $p < 0.001$) indicate that loyalty points deviate substantially from a normal distribution, with heavy tails driven by a small number of customers accumulating disproportionately high loyalty values.

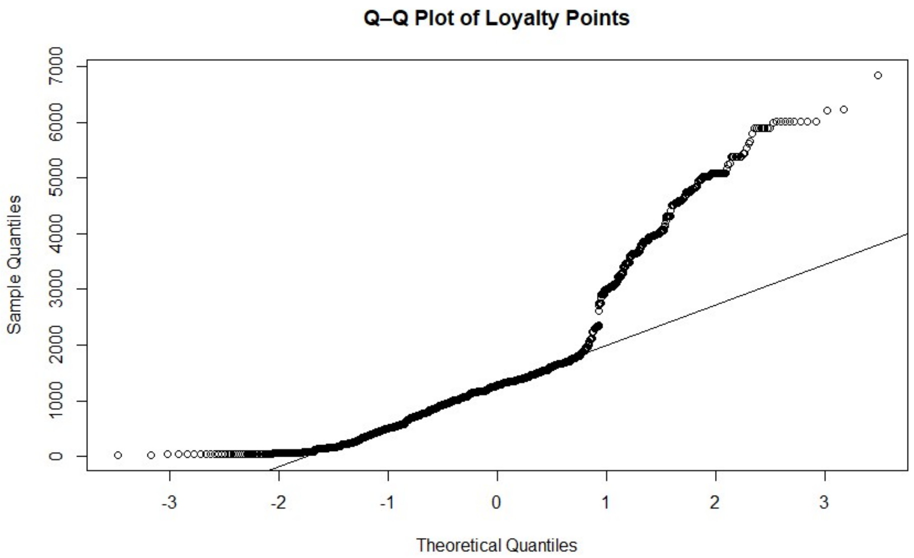

Figure 14 - Q-Q Plot of Loyalty Points

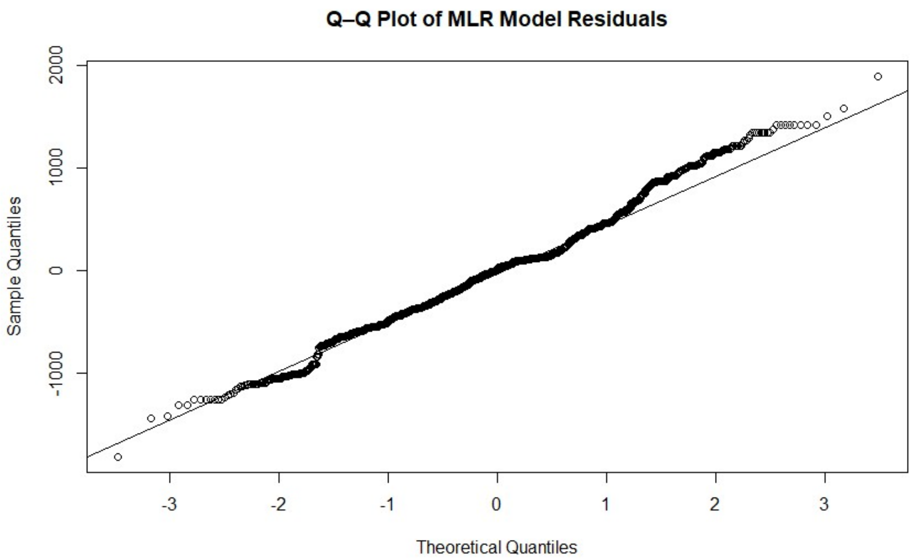

Figure 15 - Q-Q Plot of MLR Model Residuals

The regression model provides a strong and reliable view of overall loyalty behaviour, explaining approximately 84% of the variation in loyalty points. Spending behaviour and income are the primary drivers (both coefficients ≈ 34), while age plays a much smaller role (≈ 11). Diagnostic checks confirm that the model is robust for understanding population-level trends, even if individual extreme cases behave differently.

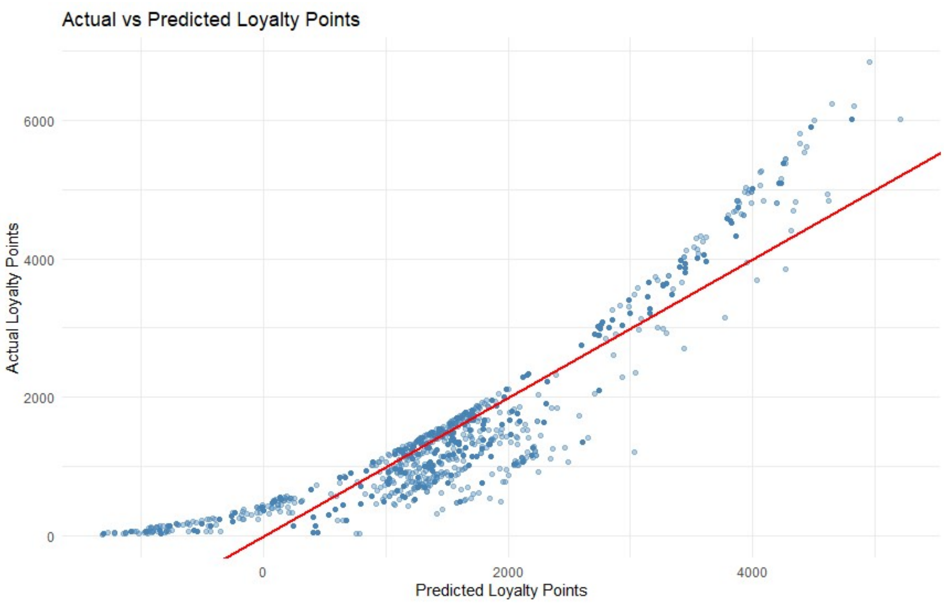

Figure 16 - Linear Regression Model

Collectively, these visual and statistical results provide a robust, evidence-based foundation for improving Turtle Games's loyalty programme and marketing strategy.

Patterns and predictions

Several consistent patterns emerged across the analysis. Loyalty point accumulation is primarily behaviour-driven, with spending score acting as the strongest predictor across all models. Remuneration plays a supporting role by enabling spending, while age has a comparatively minor effect once other variables are controlled for. The loyalty programme effectively rewards engagement but produces uneven outcomes, with a small subset of customers accumulating disproportionately high loyalty points. This is reflected in the strongly right-skewed distribution of loyalty points, with normality formally rejected (Shapiro–Wilk $W \approx 0.84$, $p < 0.001$).

Despite this skewness, the multiple linear regression (MLR) model provided strong population-level insight. The final model achieved a high goodness of fit (adjusted $R^2 \approx 0.84$; residual standard error ≈ 514). Spending score and remuneration were both highly influential, each contributing approximately 34 loyalty points per unit increase, while age contributed around 11 points per year. Residual diagnostics showed substantially improved normality, supporting the model's suitability for prediction.

Scenario predictions demonstrated that increases in spending behaviour lead to substantial gains in expected loyalty points, reinforcing the importance of incentivising engagement. However, unrealistic predictions at very low values highlight the limits of linear extrapolation.

From a business perspective, Turtle Games should prioritise encouraging incremental spend among mid-tier customers, protect high-value segments, and address usability and value concerns highlighted in customer reviews to strengthen long-term loyalty.